

Molecular dynamics modeling of tubulin C-terminal tail interactions with the microtubule surface

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ABSTRACT

Tubulin, an α/β heterodimer, has had most of its 3D structure analyzed; however, the carboxy (C)-termini remain elusive. Importantly, the C-termini play critical roles in regulating microtubule structure and function. They are sites of most of the post-translational modifications of tubulin and interaction sites with molecular motors and microtubule-associated proteins. Simulated annealing was used in our molecular dynamics modeling to predict the interactions of the C-terminal tails with the tubulin dimer. We examined differences in their flexibility, interactions with the body of tubulin, and the existence of structural motifs. We found that the α -tubulin tail interacts with the H11 helix of β -tubulin, and the β -tubulin tail interacts with the H11 helix of α -tubulin. Tail domains and H10/B9 loops interact with each other and compete for interactions with positively-charged residues of the H11 helix on the neighboring monomer. In a simulation in which α -tubulin's H10/B9 loop switches on sub-nanosecond intervals between interactions with the C-terminal tail of α -tubulin and the H11 helix of β -tubulin, the intermediate domain of α -tubulin showed more fluctuations compared to those in the other simulations, indicating that tail domains may cause shifts in the position of this domain. This suggests that C-termini may affect the conformation of the tubulin dimer which may explain their essential function in microtubule formation and effects on ligand binding to microtubules. Our modeling also provides evidence for a disordered-helical/helical double-state system of the T3/H3 region of the microtubule, which could be linked to depolymerization following GTP hydrolysis.

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INTRODUCTION

The tubulin protein polymerizes to form microtubules (MTs) which are cylindrical structures consisting typically of 13 protofilaments, each of which is comprised of repeats of α - β tubulin hetero-dimers. The carboxy (C)-terminal tail domains of all tubulin monomers are found on the outer surface of the MT. Both α - and β -tubulin consist of numerous isotypes, differing in amino acid sequence and cellular and tissue distribution.¹ The lengths and sequences of the C-terminal tails vary greatly depending on isotype; human isotypes typically consist of 10–13 residues on α -tubulin, and 18–24 residues on β -tubulin, and have a high proportion of negatively charged residues.^{2,3} A great deal of evidence indicates that the C-terminal tails of both α - and β -tubulin play critical roles in regulating MT assembly and function, and in determining the conformation of the tubulin molecule,^{4–8} roles whose details are very poorly known. C-terminal tail residues are necessary for the assembly of normal MTs; one recent cloning study has demonstrated that the first two tail residues of β -tubulin are indispensable for the formation of MTs in HeLa cells.⁴ Tubulin polymers assembled *in vitro* from tubulin lacking both C-terminal tails (denoted as $\alpha_s\beta_s$ tubulin) display an altered morphology characterized by bundles of twisted filaments, sheets,⁵ and hooks.^{6,7} Moreover, the presence of alkali metal cations⁸ or cleavage of tails⁵ allows polymers to form at lower tubulin concentrations, and the reduction of negative charges on C-termini by covalent modification lowers the rate constant for MT disassembly.⁹ Tubulin polymers lacking only the β -tubulin tail ($\alpha\beta_s$ tubulin) display a ring con-

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formation, do not require GTP for polymerization, and are temperature insensitive.¹⁰

It is likely that the effects of the C-termini on MT assembly and tubulin conformation are critically dependent on the conformational interaction between the C-termini and other parts of the tubulin molecule. Nevertheless, the natures of these interactions are almost entirely unknown. Although a series of studies have determined the structure of the rest of the tubulin molecule, the structures of the C-termini have remained largely invisible to crystallographic methods. Thus, a molecular-level picture of the structures of tubulin protofilaments obtained by molecular dynamics (MD) simulations, including both tail dynamics and their interactions with the remainder of the protein, offers an opportunity to provide unique insights into a better comprehension of the biological functions of tail domains and their significance for MTs.

Depending on the isotype, the charge of the C-terminal tail of human β -tubulin is generally between -10 and $-12 e$ and that of the α -tubulin tail is -7 or $-8 e$, not including post-translational modifications.^{2,3} Although the remainder of the tubulin dimer is also negatively charged overall (the total charge of the dimer is approximately $-50 e$), there are also positively charged regions on the protein surface. By using reductive methylation, residue Lys 394, which together with Arg 390 and His 393 forms a positively-charged pocket on the surface of α -tubulin, has been shown to be essential to MT polymerization.^{11,12} Because Lys 394 is less accessible to methylation in the MT polymer compared to when present in the dimer,^{12,13} it has been suggested that this residue is involved in electrostatic interactions with the negatively-charged acidic residues in the tail regions.¹⁴ The existence of such a “tail-body” electrostatic interaction was also supported by a study¹⁵ in which the detached α -tubulin tail was computationally docked to the tubulin surface and found to bind at a site near Lys 394. Experimental findings that C-terminal tail cleavage can affect ligands that bind to tubulin, and that this influence seems to be due to electrostatic interactions, also indicate that tail-body interactions contribute to tubulin conformation. This evidence includes a study showing that cleaving the C-terminal tail on α -tubulin lowers the activation energy for colchicine binding to tubulin, and leads to pH independent binding.¹⁶ Another such study showed that the influence of oligoanions on the stabilization of tubulin polymers by vinblastine is eliminated by the cleavage of the C-terminal tails of β -tubulin.¹⁷

The objective of this article is to use MD modeling in order to elucidate the issues surrounding the role of C-termini in structural and functional effects on the tubulin dimer and, consequently, in the biological behavior of MTs. The specific modeling tool that we apply to our analysis is simulated annealing in conjunction with MD

simulation. Simulated annealing is an enhanced sampling technique in which heating followed by re-cooling of a molecular system can be used to determine its equilibrium structure. Heating allows the system to overcome energy barriers that would otherwise prevent the system from fully sampling configuration space, and slow cooling causes the system to freeze into a minimum energy structure. This method has frequently been applied within Monte Carlo simulations to determine conformational free energy minima^{18,19}; more recently simulated annealing has been used within MD simulations,²⁰ as it is in the present work. Here, this technique is applied to obtain positions of C-terminal tails for the tubulin dimer. Subsequently, simulations are run of protofilaments formed from $\alpha\beta$ tubulin, $\alpha\beta_s$ tubulin, and from $\alpha_s\beta_s$ tubulin. In this way, the effects of the tails on protofilament dynamics are investigated computationally.

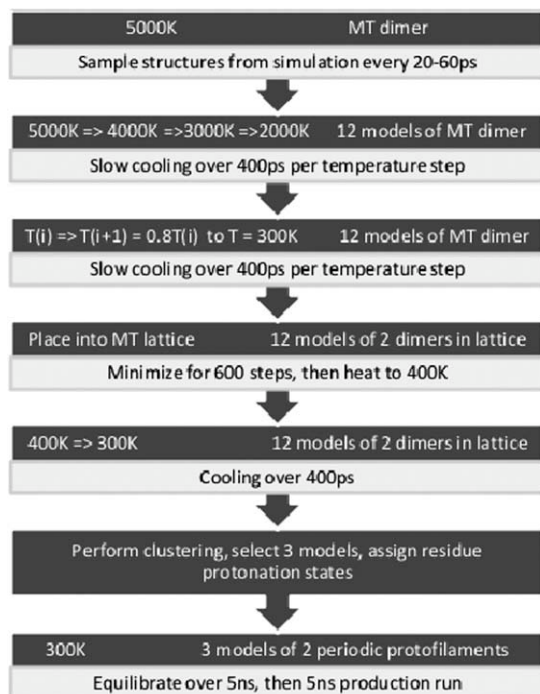
We analyze MD trajectories to investigate the occurrence of tail-body interactions of the C-terminal tail regions with positively-charged residues on the tubulin surface. Structural motifs intrinsic to the C-terminal tail domains seen in the MD trajectories are also considered. Finally, we analyze the root mean square fluctuations of residues in trajectories with and without tail domains. We show that interactions of the tail domains with exposed residues on the surface of the remainder of the protein may play an important role in MT dynamics by influencing the large-scale conformation of tubulin's intermediate domain.

MATERIAL AND METHODS

Simulated annealing

After heating the tubulin dimer system to 5000 K, each of 12 models was slowly cooled to 300 K over 5.2 ns of simulation time. Out of these 12, three models were selected as starting positions in simulations of two explicitly-solvated tubulin protofilaments. The flowchart in Figure 1 summarizes the simulated annealing/MD simulation procedure adopted in our study to determine tubulin tail conformations, which we now describe in detail.

Our system consisted of a tubulin dimer, composed of human isotypes α IV and β I, modeled as described elsewhere²¹; these have C-terminal tails with lengths of 10 and 18 residues (residues α 439–448 and β 427–444), and charges of -8 and $-12 e$, respectively. Initial set-up was performed in the *leap* and *antechamber* modules of the AMBER9²² program package using the AMBER03 force field to parameterize the tubulin protein.^{23,24} MD was performed using the *sander* module of AMBER9.²² Twelve high temperature implicitly-solvated MT dimers were generated by heating the dimer to 5000 K (chosen to make the tails mobile on a reasonable time scale), and then extracting structures from the simulation every

**Figure 1**

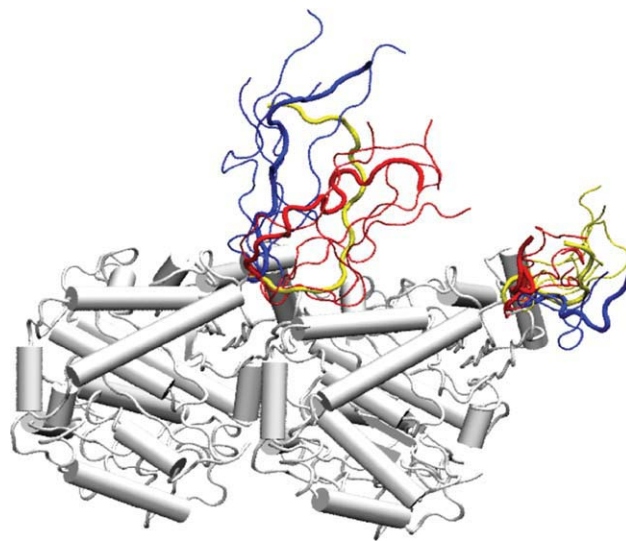
Flowchart illustrating the simulated annealing/MD simulation procedure used in this work to determine tubulin tail conformations.

20–60 ps. These were then slowly cooled to a target temperature of 300 K. Our annealing schedule used temperature reduction steps of 1000 K over 200 ps time periods, with equilibrations being performed at each of these temperatures for a period of another 200 ps, until the temperature was below 2000 K, and then used a 0.8-power law cooling-schedule again with 400 ps spent for each temperature stage. A time step of 1 fs was used throughout all simulations. The modified generalized Born method GB^{OPC} corresponding to model II in Ref. 25 (igb = 5) was employed to represent a 150 mM ionic solution and the temperature was maintained by a Langevin thermostat with a collision frequency of 2 ps⁻¹. All bonds involving hydrogens were constrained using the Shake²⁶ algorithm. A restraint weight of 15 kcal/mol was placed on residues 1–438 of α tubulin and 1–428 of β tubulin. Removal of center-of-mass translational and rotational motion was disabled. The non-bonded cut-off used was 50 Å.

Structures were fit to the original homology model by running simulations for 5000 steps with a restraint of 50 kcal/mol on the α -carbons of all residues not belonging to the C-terminal tail domains, forcing the α -carbons belonging to the main body of the dimer to assume a structure similar to the one determined by electron crystallography experiments. In some cases, the *autosculpt* feature of the Pymol program²⁷ was employed to slightly

adjust the positions of β tubulin tail residues which overlapped with parts of the opposite β tubulin. Tubulin dimers were then placed into a configuration consistent with MT geometry by using the model of Li *et al.*²⁸ Two adjacent dimers were created as would exist in the MT lattice by using an in-house VMD²⁹ script, as described elsewhere.³⁰ These were then relaxed for 300 steps without solvent, and another 300 steps with implicit solvent. Afterwards the structures were heated to 400 K, and then cooled to 300 K over a period of 400 ps for further energy minimization, with the same restraint on the α -carbons belonging to the main body.

The 12 models of the tubulin dimer resulting from our simulated annealing procedure are shown in Figure 2. Structures were clustered in the *ptraj* module of AMBER using the means algorithm³¹ to produce three clusters each for α - and β tubulin tails, by comparing positions of alpha-carbons based on root-mean-squared distance (RMSD) after fitting all tubulin dimers, excluding tail residues, to the position of one of the dimers. Clusters of α tubulin tail peptides contained five, four, and three elements, respectively, with mass weighted root-mean-square distances to the centroid of 0.6 Å for each of the three clusters. Clustering the β tubulin tail peptides resulted in one group of six elements, with a mass weighted root-mean-square distance to centroid of 1.4 Å, a second cluster with five elements and a mass weighted root-mean-square distance to centroid of 1.3 Å, and a third cluster with a single element only. Cluster

**Figure 2**

Tubulin dimer, showing three clusters obtained by simulated annealing MD for the C-terminal tail domains, colored in yellow, red, and blue, respectively. Cluster representatives *p1* (yellow), *p2* (red), and *p3* (blue) are depicted as thicker coils compared to other cluster elements. The dimer is viewed from above if positioned in a MT lattice with β -tubulin on the left and α -tubulin on the right.

representatives were chosen as the dimers having β tubulin tails closest to the cluster centroids. Because each of these three dimers fell into a different one of the three clusters corresponding to the α tubulin tails, we chose to retain the original pairing between α - and β tubulin monomers, rather than pairing cluster representatives for β tubulin tails with similarly chosen cluster representatives for α tubulin tails. As a further check on the appropriate number of representatives, hierarchical clustering of both β - and α -tubulin tails was performed using the MMTSB Toolset³² command *jclust* with the *rmsd* mode, requesting a maximum of 50 structures, which gave two main clusters for the α -tubulin tail and three for the β -tubulin tail.

The three representative dimers, which were used to perform MD simulations, are shown in Figure 2. In the text that follows, we refer to these three models as *p1*, *p2*, and *p3*. In addition, we performed simulations with one or both tail domains removed. The α -tubulin tail position for the $\alpha\beta_s$ model was taken from model *p3*.

MD simulations

The *propka* program^{33,34} was used to assign protonation states to Asp, Glu, His, and Lys residues on each of the three models; as a result His 309 was protonated on α -tubulin of model *p1*, His 307 on β -tubulin of models *p2* and *p3*, and α -tubulin residue Asp 46 was protonated on all three models.

The tubulin dimers were then placed into a MT lattice. The positions of GDP at the exchangeable site and GTP at the non-exchangeable site were taken from the PDB file 1TVK³⁵ and those of the Mg^{2+} ions from the PDB file 1JFF.³⁶ Parameterization of nucleotide co-factors was taken from reference.³⁷ The MD simulations included two adjacent protofilaments composed of two tubulin heterodimers; identical tail positions were used on both protofilaments. One hundred three Na ions were added to neutralize the system electrostatically, and following this 80 each of Na^+ and Cl^- were added to bring the ionic concentration to 100 mM. The system was placed into a periodic box with dimensions 81.2 Å by 107.5 Å by 152.5 Å, which was filled with pre-equilibrated explicit TIP3P water molecules³⁸ using the *leap* module of AMBER9. The protofilaments were aligned with the *x*-axis of the periodic box, and by setting the *x*-dimension of the box equal to 81.2 Å, or the length of one tubulin dimer, the periodic boundary conditions were taken advantage of to model the system as two periodic protofilaments. Structures were then converted from AMBER to Gromacs-style coordinate and topology formats using the *amb2gmx* perl script,³⁹ and simulations were performed with the Gromacs v3.4^{40,41} program.

Structures were energy minimized, which also relieved any steric clashes with waters. The entire system was relaxed using 5000 steps of steepest descent (1000 steps),

followed by conjugate gradient minimization. Following equilibration for 5 ns with a time step of 2 fs, MD simulation was performed over a period of another 5 ns with the same time step, with periodic boundary conditions at a temperature of 300 K and at constant atmospheric pressure. The Shake algorithm²⁶ was used to constrain bonds involving hydrogen in all simulations and we used a non-bonded cut-off of 12 Å. PME was employed to calculate long-range electrostatics.

Analysis

Analysis was then performed on the 5 ns production MD runs by using the suite of tools available in the Gromacs program package.^{40,41} Only tail domains belonging to the protofilament positioned on the left when viewed from the minus to the plus end was used for analysis. The restriction of our analysis to these tail domains was made because of their physically central position relative to our models, in contrast to the peripheral position of the other set of tail domains, as seen in Figure 5.

For the examination of structural motifs, a matrix of scaled rho (rhosc) values was computed using the Gromacs^{40,41} tool *g_rms*. This program compares each structure in trajectory *p1* to each structure in trajectory *p3*, after fitting the structures on top of one another by rotation and translation.⁴² The structural similarity value rho is defined as a function of the two structures A and B by:

$$\rho(A, B) = \frac{2\text{RMSD}(A, B)}{\sqrt{R_{\text{gyr}}^2(A)R_{\text{gyr}}^2(B) - \text{RMSD}^2(A, B)}},$$

where R_{gyr} is the radius of gyration of a structure, and where RMSD gives the root mean square deviation between two structures. Scaling the aligned principal moments of inertia of the two structures A and B to values that are all equal and satisfy $R_{\text{gyr}} = 1$ prior to comparing structures, leads to a more universal, size-independent measure of similarity, rhosc, assuming values between 0 and 2. Values of rhosc less than 0.3 were used here as an indication of structural similarity, and the percentage of comparisons for which $\text{rhosc} < 0.3$ represents the percentage of total time for which the conformation of the motif is similar between the two models A and B.

The *g_min_dist* tool was employed to determine minimum distances between charged or aromatic residues belong to tubulin tail domains and other charged or aromatic residues. Residue pairs were included in this analysis if they came closer than 0.3 nm at any point during the simulations.

Average root-mean-square fluctuations (RMSF) were calculated using the *g_rmsf* program, as the averages over the simulation time, $\langle |\Delta R_i|^2 \rangle$, in the square fluctuation of

Table I

Charged and Aromatic Residues Involved in Contacts Between α -Tubulin's C-Terminal Tail Domain, α -Tubulin's H10/B9 Loop, and β -Tubulin's H11 Helix

	α tail- β H11	β H11- α H10/B9loop	α tail- α H10/B9loop
<i>p1</i>	α Tyr440,Glu441- β Arg390 α Asp442,Glu443- β Arg391	β Arg390- α Asp345	α Tyr440- α Asp345
<i>p2</i>	α Tyr440- β Arg390 α Tyr440- β Arg391	β Arg391- α Trp346	
<i>p3</i>	α Tyr440- β Arg390	β Arg390- α Asp345 β Arg391- α Trp346	α Tyr440- α Asp345
$\alpha\beta_S$	α Glu445,Glu447,Glu448- β Lys379 α Asp442- β Arg390		α Tyr440- α Asp345
$\alpha_S\beta_S$		β Arg390,Arg391- α Trp346	

the position R_i of the α -carbon corresponding to each residue i .

Two programs *do_dssp* and *g_helix* were used in the analysis of helical content. The *do_dssp* program assigns secondary structure by identifying patterns of hydrogen bonding using the DSSP⁴³ algorithm. On the other hand, *g_helix* determines helicity based upon both hydrogen bonds and ϕ/ψ angles.

RESULTS AND DISCUSSION

Simulated annealing

Following annealing, α -tubulin tails were generally found to point in the longitudinal direction toward where the β -tubulin of the neighboring dimer would be in a MT lattice (see Fig. 2). There was a tendency of the β -tubulin tails to point laterally in the direction of where the right-hand adjacent protofilament would be if the dimer were part of a MT being viewed from the minus to the plus end (i.e., towards the intermediate, taxol-binding domain). However, the β -tubulin tails covered a large range of conformational space (Fig. 2).

By clustering the final endpoints of simulated annealing, and choosing representatives of different clusters (*p1*, *p2*, and *p3*) for subsequent simulation and analysis, we ensured that these three simulations were run from well-separated initial positions. Thus, while each of these simulations on its own does not demonstrate complete time-sampling of MD, a broader picture of the dynamic behavior of the protein system was obtained by considering all three simulations.

Structural motifs

A previous set of replica exchange MD simulations studied peptides composed of the residues forming the C-terminal tails of β -tubulin and identified three transient structural motifs as potential sites for casein kinase-2 (CK-2) phosphorylation and MAP2 binding.³ We sought to determine, by analysis of our trajectories, whether these structural motifs found for the isolated tail

structures, still characterize the tail domains within the context of a MT.

The proposed CK-2 motif, consisting of the residues DATA, occurs at the start of the β tubulin tail and is evolutionarily conserved between tubulin isoforms.³ The structure of these four residues beginning with residue Asp 427 may be seen in Figure 3. The percentage of total comparisons for which rhosc values are less than 0.3 was found to be 93.6%, indicating a high degree of structural similarity for this sequence and is higher than that found in the original study.

Two motifs on the C-terminal tail, that were previously identified as making up the MAP2 binding site,⁴⁴ consist of sequences EEAE (residues 438–441) and EEEA (residues 441–444), respectively, contained in the last seven residues (EEAEAAA) of the β -tubulin tail domain.³ The respective counts of rhosc values less than 0.3 for these two sequences are 14.7% and 2.8% of total comparisons. This is considerably less than in the previous study and our analysis suggests that these two regions take on little specific structure; however, the bound β -tubulin tail conformers identified below indicate that these residues remain solvent exposed and the possibility of a MAP2 induced structure cannot be ruled out.

Tail-body interactions

We performed analysis on our MD trajectories to investigate the occurrence of tail-body interactions of the C-terminal tail regions with positively-charged residues on the tubulin surface. Tables I and II show the charged residues that, at any point during our simulations, are involved in close contacts of oppositely-charged side-chains, with inter-residue distances of less than 0.3 nm, for each model. Because there were some very noticeable interactions between aromatic residues from the tails and charged residues from the tubulin body and vice versa, these were also included in our analysis. Figures 3 and 4 display models highlighting the tubulin residues involved in tail-body interactions of the β - and α -tubulin tails, respectively, for any one of the four models *p1*, *p2*, *p3*, and $\alpha\beta_S$.

Table II

Charged and Aromatic Residues Involved in Contacts Between β -Tubulin's C-Terminal Tail Domain, β -Tubulin's H10/B9 Loop, and α -Tubulin's H11 Helix

	β tail- α H11	α H11- β H10/B9loop	β tail- β H10/B9loop
<i>p1</i>	β Glu431,Glu432- α His393 β Glu432- α Lys394	α Lys394- β Glu343 α His393, Lys394- β Trp344	β Glu431- β Trp344
<i>p2</i>	β Glu432- α Lys401		
<i>p3</i>		α Lys401- β Trp344	β Asp427- β Trp344
$\alpha\beta_s$		α Arg390,His393,Lys394- β Glu343 α Lys401- β Trp344	
$\alpha_s\beta_s$		α Lys401- β Trp344	

We observed that two kinds of tail-body interactions could take place. First, charged Glu and Asp residues of the C-terminal tails can interact with a cluster of positively-charged residues on H11 of the neighboring tubulin dimer. That is, the α -tubulin tail interacts with the H11 helix of β -tubulin, and the β -tubulin tail interacts with the H11 helix of α -tubulin. Figure 5 shows the positions of the H11 domain within the MT lattice, as well as the positions of the other domains that will be discussed below. At the inter-dimer longitudinal interface,

negatively-charged residues of the α -tubulin tail interact with residues Lys 379, Arg 390, and Arg 391 on the H11 helix of β -tubulin. Likewise, at the intra-dimer interface, the β -tubulin tail interacts with the positively-charged cluster on α -tubulin containing the residue Lys 394, which has previously been identified as essential for MT polymerization, and His 393; Lys 401 on H11 can also be involved in this type of interaction. The charged tail residues involved are 441–443, 445, and 447–448 on α -tubulin and 431–432 on β -tubulin. We also noted that the ar-

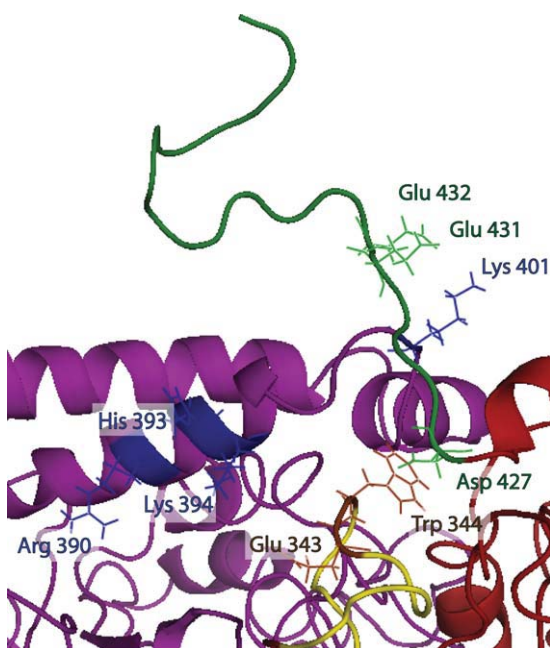
**Figure 3**

Illustration of the intra-dimer longitudinal interface within a MT between α -tubulin (purple) and β -tubulin (red). Charged and aromatic residues have been labeled and represented in stick mode that are involved in close contacts between the β -tubulin tail, shown in green cartoon, and either the β -tubulin H10/B9 loop, shown in yellow cartoon, or the α -tubulin H11 helix, as well as those involved in interactions between the latter two regions. This figure was created using the Pymol molecular viewing program,²⁷ using model *p2*.

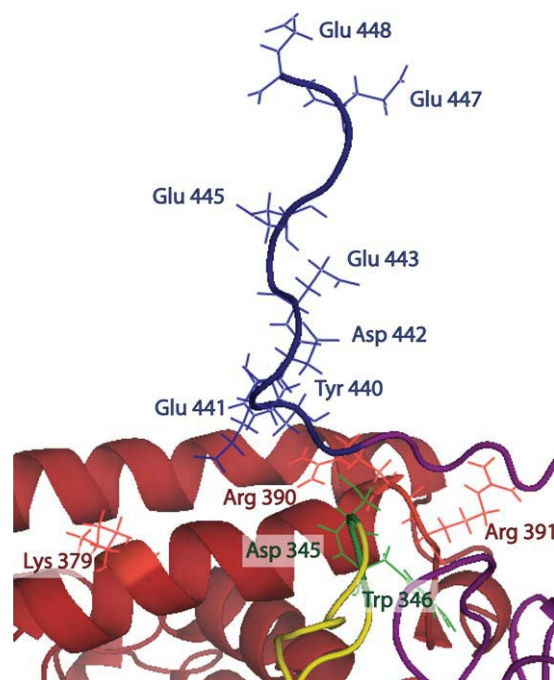
**Figure 4**

Illustration of the inter-dimer longitudinal interface within a MT between α -tubulin (purple) and β -tubulin (red). Charged and aromatic residues have been labeled and represented in stick mode that are involved in close contacts between the α -tubulin tail, shown in blue cartoon, and either the α -tubulin H10/B9 loop, shown in yellow cartoon, or the β -tubulin H11 helix, as well as those involved in interactions between the latter two regions. This figure was created using the Pymol molecular viewing program,²⁷ using model *p3*.

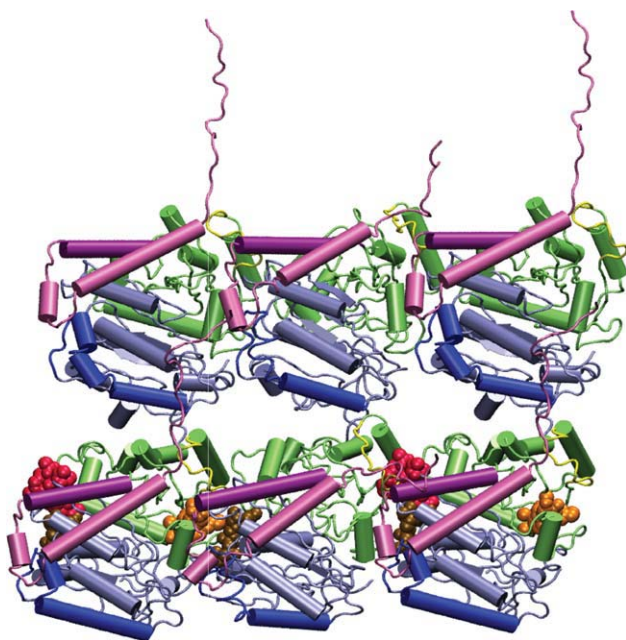


Figure 5

Representation of part of the MT lattice, showing β - α tubulin monomers comprising each of two protofilaments oriented with plus ends on the left-hand side. N-terminal domains are colored in blue, intermediate domains in green, and C-terminal domains in purple. The T3/H3 region is shaded in dark blue, the H10/B9 loop in yellow, and the H11 helix in dark purple. The positions of the binding sites of colchicine (orange), vinblastine (red), and nucleotides GDP and GTP (brown) are also illustrated for one of the protofilaments.

omatic residue Tyr 440 on the α -tubulin tail interacts with charged residues Arg 390 and Arg 391 on the H11 helix of β -tubulin by nonpolar side-chain stacking, as well as by interactions between charged Arg side-chains and the polar hydroxyl substituent on the Tyr ring.

The second kind of tail-body interaction observed was between tail residues and residues on the H10/B9 loop of the same tubulin monomer. These contacts are not between two charged residues but rather involve an aromatic residue and a charged residue. In the case of α -tubulin, the tail residue Tyr 440 interacts with Asp 345 of the H10/B9 loop; this contact involves both hydrophobic interactions between the two bulky side chains, and electrostatic attraction between the charged carboxylic acid group of the Asp side-chain and the polar hydroxyl substituent on the Tyr ring. In the case of β -tubulin, tail residues Asp 427 or Glu 431 engage in close nonpolar contacts with Trp 344 of the H10/B9 loop.

Interestingly, the H10/B9 loop can also interact with the positively-charged cluster on the H11 helix of the neighboring dimer, and these interactions are also shown in Tables I and II. In the case of the α H10/B9 loop and the β H11 helix, there can be a salt bridge established between α Asp 345 and residue Arg 390 of H11; also the bulky aromatic side chain of α Trp 346 engages in

non-polar interactions with the long side chains of Arg 390 or Arg 391 from β H11. In the case of interaction between β -tubulin's H10/B9 loop and α -tubulin's H11 residues, Glu 343 of the β -tubulin loop can interact with Arg 390, His 393, and/or Lys 394 on α -tubulin's H11. Here also β Trp 344 interacts with residues His 393, Lys 394, and/or Lys 401 from the H11 helix of α -tubulin through both non-polar side-chain interactions and contacts between the aromatic rings of Trp and the positively-charged amine group on Lys. Note that it is likely that these (and possibly some of the other instances of interactions between positively-charged and aromatic residues discussed above) are cation- π interactions, although our simulations did not employ a polarizable force-field capable of representing the electronic nature of these interactions.

For model *p1*, we plot in Figure 6 how minimum distances involving tail-body interactions of the α -tubulin tail vary throughout the simulation. During the first half of this simulation there is a strong interaction between β -tubulin's helix H11 and the H10/B9 loop of α -tubulin, at the longitudinal inter-dimer interface. During the second half of the simulation these interactions are reduced and instead increased interactions are observed between the α -tubulin tail and both β -tubulin's helix H11 and the H10/B9 loop of α -tubulin. This switch in interactions between the C-terminal tail, the H10/B9 loop, and the H11 helix of the neighboring dimer may be indicative of competing interactions between these three regions.

At the intra-dimer interface, as seen in Supporting Information Figure S3, for the *p1* model there are multiple close contacts between the β -tubulin tail, α -tubulin's helix H11, and the H10/B9 loop of α -tubulin. Although, as may be seen from Tables I and II, many residues or pairs of residues involved in tail-body interactions and in H11-H10/B9 loop interactions are shared between models *p1*, *p2*, and *p3*, these models explore different regions in conformational space and thus their trajectories display qualitatively different behaviors. For example, in model *p2*, at the intra-dimer interface, the β -tubulin tail interacts with residue Lys 401 of the H11 helix on α -tubulin, but there is no interaction between α H11 and the H10/B9 loop of β -tubulin. On the other hand, in both the *p3* and $\alpha\beta_S$ simulations, α H11 interacts with β -tubulin's H10/B9 loop but not with the β -tubulin tail. In the $\alpha_S\beta_S$ simulation, without either tail domain present, almost continuous interaction was observed between the H10/B9 loop and the H11 helix of the adjacent monomer, at both monomer interfaces (see Supporting Information Fig. S2).

A physical model was previously created of ion dynamics in and around MTs, based on the application of a random alternating electric force on the surrounding cations by fluctuations in C-terminal tail positions.⁴⁵ The continuous breaking and formation of salt bridges give rise to an alternating voltage source on ions as the

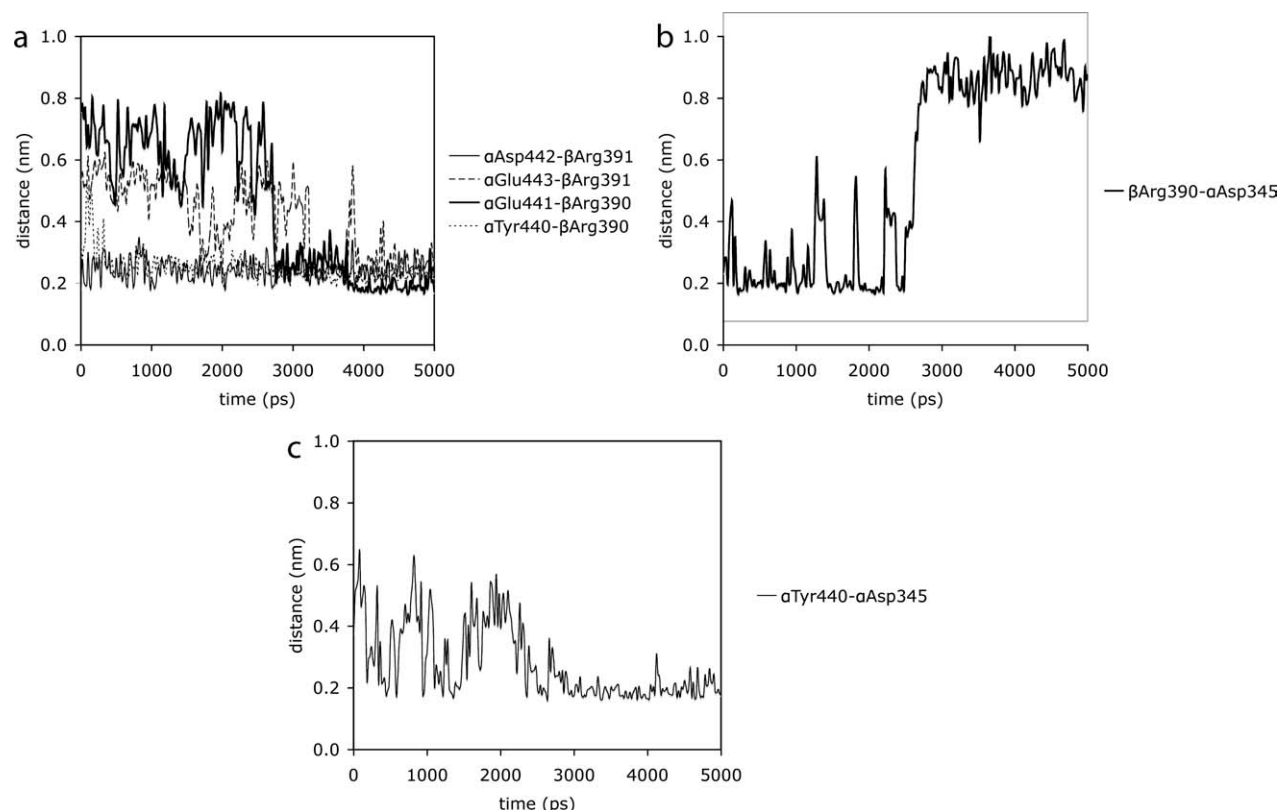


Figure 6

Minimum distances (nm), plotted for the *p1* model as a function of simulation time (ps), between charged or aromatic residues belonging to (a) α -tubulin's C-terminal tail domain and β -tubulin's H11 helix; (b) β -tubulin's H11 helix and α -tubulin's H10/B9 loop; and (c) α -tubulin's C-terminal tail domain and α -tubulin's H10/B9 loop.

tails dynamically oscillate between conformations where they either interact with the tubulin body or extend into more solvated conformations, as seen in Figure 6. This effect observed in our simulations appears to support the hypothesis that local charge reconfiguration due to the motion of C-termini tails can affect ionic flows and hence endow MTs with unique electro-conformational properties.

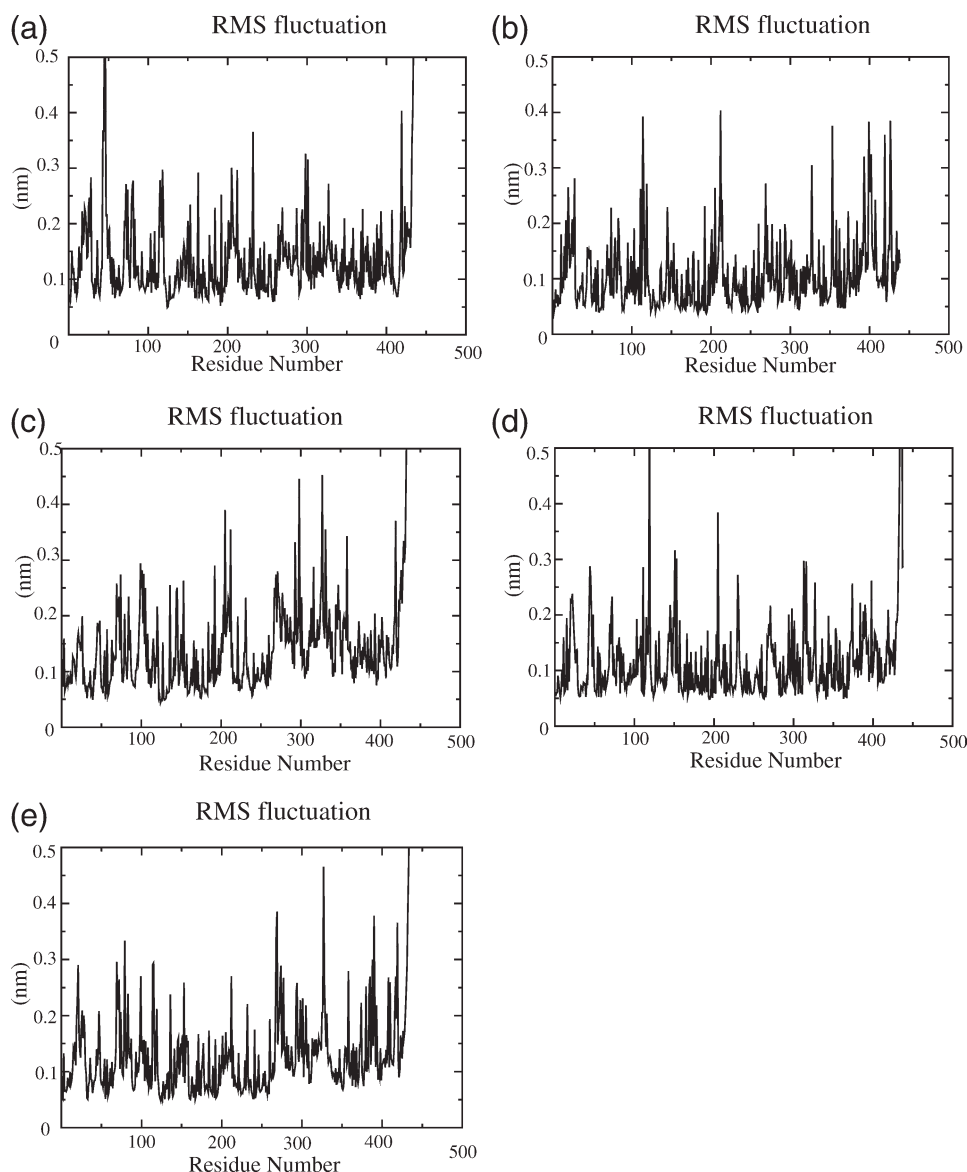
Structure and flexibility of tubulin body

The average RMSF of each residue of α -tubulin (excluding tail-residues) is plotted as a function of residue index *i* in Figure 7, and that of β -tubulin in Supporting Information Figure S6, corresponding to each of our simulations. The peaks in the RMSF plots show regions of high flexibility, and the highest peaks generally occur at solvent-accessible loops. Tubulin is known to have three main domains, which have been shown to be associated with major correlations in structural fluctuations.⁴⁶ These are the N-terminal nucleotide-binding domain, the intermediate taxol-binding domain, and the C-terminal domain, and consist of residues 1–205, 206–381,

and 382–448, respectively, on α -tubulin and residues 1–203, 204–371, and 372–444, respectively, on β -tubulin. The last few residues of C-terminal domains show a high degree of flexibility in all models; the loop connecting helices H11 and H12, and H11 itself, may also be seen from the RMSF plots to be quite flexible for the *p3* and $\alpha_5\beta_5$ models. We will now briefly examine the main characteristics of RMSF plots for α - and β -tubulin that we observed first for the N-terminal domain, and then for the intermediate domain; we will also discuss some notable differences between models in RMS fluctuations and their significance.

N-terminal domain (residues α 1-205, β 1-203)

In the N-terminal domain, peaks are seen in the RMSF plots (Fig. 7 and Supporting Information Fig. S6) at residues α 43–63, β 45–47 in the region of the H1/B2 loop, which forms lateral contacts with neighboring monomers; at residues α 69–74, β 66–71 corresponding to the T2 loop; in the region of the H3 helix and T3 loop at residues α 95–119, β 91–120; and at residues α 141–146 in the region of the glycine-rich signature loop, T4. In

**Figure 7**

Time-averaged RMSF (nm) of each residue of α -tubulin plotted as a function of residue number for model (a) $\alpha\beta_S$, (b) $\alpha_S\beta_S$, (c) $p1$, (d) $p2$, and (e) $p3$.

Figure 7 and Supporting Information Figure S6, it may be noted that the peaks corresponding to the T3/H3 region on both α - and β -tubulin are much more pronounced in the RMSF plots for model $p2$ than in the other plots. The T3/H3 region (residues $\alpha 97$ – 128 and $\beta 95$ – 126) is located close to the nucleotide binding site, the exchangeable one in the case of β -tubulin, and the non-exchangeable one for α -tubulin. When secondary structure of the N-terminal domain is compared between the simulation trajectories (see Table III and Supporting Information Table S1), it is observed that there is a loss of α -helical structure for the $p2$ trajectory compared to for the other trajectories. Upon examination, one

noticeable instance of this is a less helical appearance of the H3 helix.

We, therefore, analyzed local secondary structure of the T3 loop and of the H3 helix for each of our models. In Figure 8 and Supporting Information Figure S7, we have plotted the percentage of the total simulation time during which each residue in this region is found in a helical form, calculated by the *g_helix* analysis tool. The plots in Figure 8 show that on average during the simulations the α -tubulin T3/H3 region has lost helicity for both the $\alpha\beta_S$ model and $p2$ models, and that the β -tubulin T3/H3 region is much less helical for model $p2$ compared to those in the other models.

Table III

Percentages of Residues in α -Tubulin's N-Terminal Domain Forming Various Types of Secondary Structure, Determined Using the do_dssp Gromacs^{32,33} Analysis Tool

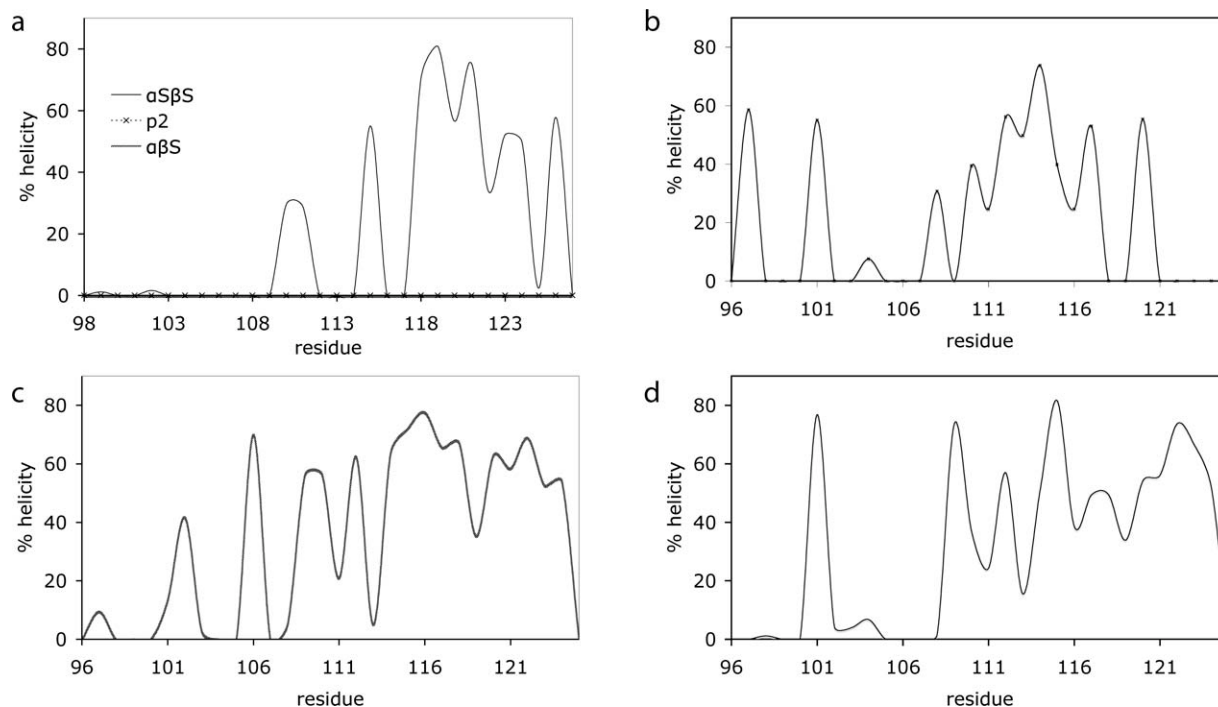
	$\alpha\beta_S$	$\alpha_S\beta_S$	$p1$	$p2$	$p3$
Structure	36	53	40	43	45
Coil	35	27	33	26	32
β -Sheet	2.2	7.4	1.9	11	6.2
β -Bridge	3.2	2.9	2.4	4.4	1.7
Bend	26	18	25	25	21
Turn	15	21	15	21	17
α -Helix	15	22	21	6.6	20
5-Helix	0.0	0.0	0.0	0.0	0.0
3-Helix	3.6	2.1	1.4	6.4	2.2

The category "Structure" represents all residues classified as α -helix, β -sheet, β -bridge, or turn.

Intermediate domain (residues $\alpha 206$ – 381 , $\beta 204$ – 371)

Major peaks seen in the RMSF plots (Fig. 7 and Supporting Information Fig. S6) in tubulin's intermediate domain are those corresponding to the region of longitudinal contacts at residues $\alpha 200$ – 232 , $\beta 201$ – 215 ; the flexible M-loop on β -tubulin at residues 276–284, and the N-loop on α -tubulin at residues 264–288; and the H10/B9 loop at residues $\alpha 339$ – 350 and $\beta 337$ – 348 . Greater RMS fluctuations of the intermediate domain of α -tubulin in

the $p1$ simulation compared to in the other simulations can be seen from the higher peaks on the RMSF plot corresponding to this simulation, and generally increased RMSF values for large parts of this domain. One instance of this is the greater fluctuation seen for model $p1$ of residues corresponding to the H10/B9 loop. This is most likely related to the interchange of interactions between the H10/B9 loop, the H11 helix of β -tubulin, and the α -tubulin tail domain, as is observed in Figure 6 for this trajectory. That is, the formation and breaking of salt bridges causes the H10/B9 loop to assume two or more different positions. These interactions may indirectly lead to higher fluctuations of other parts of the intermediate domain of α -tubulin as well. In Figure 6, the sporadic interchange of interactions between the three domains is seen to occur primarily during the first half of the simulation. We thus examined the RMSF plot calculated for the first half of the simulation only [Supporting Information Fig. S8 (a)]. This plot also shows a high degree of fluctuation of residues in tubulin's intermediate domain. However, consistent with the more constant pattern of interactions during the second half of the simulation, the RMSF plot corresponding to this time interval shows much less fluctuation of the intermediate domain [Supporting Information Fig. S8 (b)]. Notice also that models $p2$ and $\alpha_S\beta_S$ for which interactions between residues of

**Figure 8**

Percentage of total simulation time during which residues are found in a helical form plotted as a function of residue number for (a) the T3 loop/H3 helix domain of α -tubulin corresponding to models $p2$, $\alpha\beta_S$, and $\alpha_S\beta_S$; and for the same domain of β -tubulin corresponding to models (b) $p2$, (c) $\alpha\beta_S$, and (d) $\alpha_S\beta_S$.

tail domains and H10/B9 loops are entirely absent show much less fluctuation of residues in the intermediate domain than any of the other three simulations.

Experiments have shown that the polymers assembled from $\alpha_5\beta_5$ tubulin display an increased number of lateral interactions compared to normal MTs, causing junctions to form between polymer walls so that structures resemble hooked MTs with crescent-shaped cross-sections.^{6,7} Structural changes in the intermediate domains of normal MTs caused by the disruption of H10/B9 loop-H11 helix interactions by tail domains may possibly lead to changes in positions of key structural elements involved in lateral contacts, which are located in the intermediate domain, such as the M or N loop.

Relevance to drug binding and efficacy

The observations made above, from our simulations, of interactions of the tubulin C-terminal tails with the rest of the protein and our suggestion that changes in the position of tubulin's intermediate domain arising from these interactions may offer an explanation for the role of tail domains in regulating MT assembly, will now be considered in the context of previous work performed by experimentalists.

As has been mentioned in the introduction, experiments have indicated that tail-body interactions within tubulin and tubulin polymers are capable of influencing the binding or efficacy of the tubulin-targeted cancer chemotherapy drugs, colchicine and vinblastine. Colchicine and vinblastine bind to opposite ends of β -tubulin, the former at the intra-dimer interface and the latter on the other end of β -tubulin close to where the N terminus of the α -tubulin tail from the neighboring dimer interfaces with β -tubulin in the context of a protofilament⁴⁷ (see Fig. 5). On the other hand, as we will now discuss, colchicine's binding is affected by cleaving the α -tubulin tail,¹⁶ and vinblastine activity by cleaving the β -tubulin tail.¹⁷ In this connection, it is noteworthy to mention that the effect of phalloidin A on the conformational stability of tubulin is also affected by the cleavage of tail domains. Phalloidin A, a non-competitive inhibitor of vinblastine binding⁴⁸ is an extremely good inhibitor^{49,50} of the spontaneous denaturation^{51,52} to which the tubulin dimer is prone. $\alpha_5\beta_5$ tubulin is more resistant to this decay than is undigested $\alpha\beta$ tubulin, and the stabilizing effect of phalloidin A is even stronger with $\alpha_5\beta_5$ tubulin than with $\alpha\beta$ tubulin.⁵³

In experiments performed by Rai and Wolff¹⁷ it was observed that the negative effect that oligo-anions such as sodium triphosphate have on vinblastine-induced polymerization of tubulin is prevented when the C-terminal tail on β -tubulin is removed by subtilisin digestion, although subsequently removing the α -tubulin C-terminal tail to form $\alpha_5\beta_5$ tubulin has no further effect. Note that these polymers are not MTs but are filamentous

spirals.⁵⁴ Because this negative effect of oligo-anions on vinblastine-induced polymerization of tubulin could be induced in $\alpha\beta_5$ tubulin (recall that $\alpha\beta_5$ tubulin defines dimers lacking only the β -tubulin tail) by increasing the concentration of oligo-anions from 1 mM to 10 mM, a model was suggested in which there is an interaction favorable to polymer stabilization that is interfered with by both the β -tubulin tail domain and oligo-anions. More specifically, it was proposed that some residues on the tubulin body might interact with a positively-charged residue or region on the tubulin surface, and compete with β -tubulin tail residues for interactions with the same positive cluster.¹⁷ An additive negative effect on stabilization from both this tail-body interaction and the blockage of the positive region by the added anions would then provide an explanation for both how the cleavage of tails allows vinblastine to stabilize tubulin polymers in the presence of a low concentration of anions, and how, when this concentration is increased, vinblastine cannot stabilize even $\alpha\beta_5$ polymers. The interaction between the H10/B9 loop on β tubulin and the positively-charged cluster on H11 of the neighboring α -tubulin, observed in our simulations, along with our observation of interactions between the β -tubulin tail and both of these domains competing with the former interaction, support such a model. From this perspective, the interaction between the H10/B9 loop on β tubulin and positively-charged residues on H11 of the neighboring α tubulin could be strengthened by vinblastine binding, contributing to vinblastine-induced polymer stabilization. On the other hand, the addition of anions would be expected to interfere with this longitudinal interaction. However, without the additive effect of the tail-body interactions, in $\alpha\beta_5$ polymers, vinblastine (albeit while bound to the end of β -tubulin distant from the H10/B9 loop) may be capable of stabilizing either the longitudinal loop-helix interaction and/or the accompanying large-scale effect on the position of the intermediate domain, in the presence of anions. The latter effect can be overridden by adding enough anions.

The effect of vinblastine on tubulin polymerization has been the object of investigations for many years. It turns out that vinblastine binds to MT ends with high affinity but also binds with a much lower affinity to tubulin sites located along the surface of the MT cylinder.^{55–57} Approximately, 16–17 high-affinity binding sites located at MT ends have been determined to exist per MT. They are responsible for the well-known kinetic suppression effects of vinblastine, that is, its stabilization of MTs, at low concentrations (sub-micromolar).⁵⁸ The binding of vinblastine to the low affinity binding sites on the MT surface is believed to be responsible for the depolymerization of MTs at intermediate ($> 10 \mu\text{M}$) drug concentrations. Finally at drug concentrations of $> 10 \mu\text{M}$, the tubulin polymer stabilized by vinblastine, studied by Rai and Wolff,¹⁷ is an alternate form of the polymer, not the

MT polymer. One possible explanation of the opposite effects of vinblastine on MT polymerization at sub-micromolar versus intermediate concentrations may be that in the vinblastine-bound polymer, the H11-H10/B9 loop interaction is stronger than normal, which can lead to increased MT stabilization when vinblastine binds to MT tips only; increasing the strengths of these interactions throughout other parts of the MT may lead to destabilization.

In a separate set of experiments, Chakraborty *et al.*¹⁶ found that cleaving the α -tubulin tail domain lowers the activation energy and the on-rate for colchicine binding to the tubulin dimer. In contrast to the case of vinblastine effects on tubulin polymers, the effect is seen in $\alpha_5\beta_5$ tubulin but not $\alpha\beta_5$ tubulin, that is, cleavage of the β -tubulin tail alone has no effect on colchicine binding. Moreover, the binding affinity of colchicine for the tubulin dimer is normally dependent on pH, a factor known to affect tail conformation, but for the $\alpha_5\beta_5$ tubulin dimer, binding is independent of pH. To explain these findings, the authors suggested that although colchicine binds to β -tubulin at a site distant from the α -tubulin tail, tail-body interactions of the α -tubulin tail can cause large-scale conformational changes in the tubulin protein, which both promotes the binding of colchicine and leads to a higher activation energy for binding.¹⁶ In support of this hypothesis it was noted that a 100 mM concentration of added NaCl, which would partially screen the electrostatic tail-body interactions, also lowers the activation energy and the on-rate for colchicine binding. It was speculated that the tail-body interaction might involve bending of the α -tubulin tail towards the positively-charged cluster on H11 of α -tubulin, affecting the colchicine binding site and leading to the altered kinetics.^{15,16} This was not observed in any of our simulations. However, our MD modeling does support the proposed mechanism by which a tail-body interaction causes a conformational change in α -tubulin by a slightly different explanation, since our results indicate that the α -tubulin tail can interact with the H10/B9 loop, which may enable the intermediate domain to flex out laterally, leading to significant structural changes between normal and $\alpha_5\beta_5$ tubulin. By this line of reasoning, these changes would occur not only on the side of α -tubulin's intermediate domain close to the tail domain, but also extend to the distant side of this intermediate domain, near where colchicine binds to β -tubulin. It is also interesting to speculate whether, conversely, the drug activity of colchicine, which inhibits MT assembly, may involve altering the position of the intermediate domain of α -tubulin.

In summary, the combined results of the experiments from the literature,^{16,17} which we have just described, and the simulations reported in the present article seem to indicate that both vinblastine effects on tubulin polymers and colchicine binding to tubulin are affected by shifts in the position of the entire intermediate domain

of tubulin compared to that found in $\alpha_5\beta_5$ polymers or tubulin dimer, in the former case that of the β -tubulin monomer and in the latter that of α -tubulin. We propose that this shift is brought on by interactions between the tail domain, the H10/B9 loop, and in the case of vinblastine, the H11 helix of the neighboring monomer, all at the other longitudinal monomer interface from where drug-binding occurs. Experimental confirmation of large-scale conformational flexibility for tubulin's intermediate domain has already been supplied by photo-affinity labeling results demonstrating that cysteines 295, 305, 315, and 316 in the intermediate domain of α tubulin, but distant from the colchicine binding site, are affected by colchicine binding.⁵⁹

CONCLUSIONS

Using computational techniques, we have determined two modes of interaction between the C-terminal tail domains with the main body of tubulin. Specifically, we have found an inter-monomer mode between charged residues on the $\alpha(\beta)$ tail and $\beta(\alpha)$ H11 and an intra-monomer mode, pairing aromatic (charged) residues on the $\alpha(\beta)$ tail with charged (aromatic) residues on the $\alpha(\beta)$ H10/B9 loop. As one would expect from the heretofore-elusive structures of the C-terminal tails, these interactions are non-specific and transient.

However, we also observe that the binding sites for the tails may bind laterally with each other, that is, the $\alpha(\beta)$ H10/B9 loop binds to $\beta(\alpha)$ H11. The result of this competitive interference is to increase flexibility in the intermediate domain with respect to $\alpha_5\beta_5$ tubulin. Such competitive behavior, in which another region of tubulin forms electrostatic interactions with the same positively charged region that the tubulin tail also binds to, has previously been hypothesized based on experimental results implicating tail-body interactions in vinblastine effects on tubulin polymers.¹⁷ Based on the connection that we predict between flexibility of the intermediate domain and tail-body interactions, together with the experimental data showing that the binding or activity of colchicine¹⁶ and vinblastine,¹⁷ respectively, depend on electrostatic interactions of the tubulin tail domains far from the drug binding site, it seems likely that the binding or activity of these two drugs may be linked to the flexibility of tubulin's intermediate domain.

A natural test of the intermediate domain flexibility hypothesis and our proposed intra-monomer binding mode is to mutate residues Glu431 and Glu 432 belonging to the C-terminal tail domain of β -tubulin to alanines and then determine whether the negative effect of oligo-anions on vinblastine-induced polymerization of tubulin is eliminated, just as it is by cleavage of β -tubulin tail domains.¹⁷ The analogous test for the inter-monomer mode is to mutate either of the two residues

Asp α 345 on the H10/B9 loop or Tyr α 440 on the C-terminal tail, predicted here to form extensive interactions with each other, to alanine, which should lead to pH independent binding of colchicine, in a similar manner that cleavage of the α -tubulin tail domain does.¹⁶

The C-terminal tails are also involved in the binding of several MAPs essential to the function of MTs. Following Luchko *et al.*,³ we characterized three structural motifs possibly involved in the CK-2 phosphorylation and MAP2 binding. Falquet *et al.*⁶⁰ previously identified a conserved sequence for CK-2, which is found as DATA in the β I-tubulin tail. The DATA motif was found to have a very high similarity among our simulations; at least for the β I tail sequence, this indicates that the structure is persistent, and does not need to be induced for phosphorylation. As this structural motif has now been identified by two independent MD protocols, it should be considered a strong prediction. Contrary to the previous findings, the two proposed MAP2 binding motifs did not maintain similar structures between the models compared. However, since the sequence is experimentally known to be significant⁴⁴ and the residues are fully solvated, the possibility that these motifs are induced fit binding sites cannot be ruled out. Rather, further experimental and computational studies are needed to address this question. Some of this type of experimental study has been carried out on α -tubulin tails already, but we have not investigated proposed binding sites from this work.⁶¹

Beyond the scope of this study are the wide variety of post-translational modifications, particularly effecting the C-terminal tail domain, that tubulin is subject to. These include tyrosination/detyrosination of α , deglutamylation of α , polyglutamylation of both α and β , polyglycylation of both α and β , and phosphorylation of certain isoforms of β .¹ All of these could well affect the conformation of the tail domains and several also would alter the charges. These modifications may play critical roles in regulating the functions of MTs; for example, detyrosination of α promotes MT stability,⁶² while polyglycylation of β is necessary for normal axoneme formation and polyglutamylation of α and/or β appears to mediate binding of MT-associated proteins (MAPs) and the motor protein kinesin to tubulin.^{14,63–67} Some of these effects are likely to be indirect, that is, the presence of the modification in the C-terminal could conceivably act as a signal to some other protein or proteins which in turn could mediate the effect, that is, MT stability, axoneme formation, and so forth. Regardless, these modifications are of such scale that it is easy to hypothesize that they cause major conformational changes in the C-termini and that these changes in turn may mediate the ultimate effects of these modifications. It is likely that there exists a complex feedback loop between the probability of modification occurring and the conformational effect of modification. In addition, the binding of the

C-termini to MAPs and motors is critical to the function of a variety of MT populations and is likely to also be regulated by conformational interactions involving the C-termini. These conformational effects and interactions and their likely functional outcomes are currently being examined.

Finally, we wish to comment on the lack of helicity of the T3/H3 region that was noted in two of the simulations. The issue of how MTs are destabilized by GTP hydrolysis is currently highly debated.^{68–72} Along with MTs, actin filaments comprise one of three major components of the cytoskeleton and these two types of protein filaments have many remarkably similar properties including the utilization of a hydrolyzable nucleotide in polymer dynamics.⁷³ Experimental and computational studies indicate that in the case of actin polymerization, a loop-to-helix transition of the DNase-I binding (DB) loop is associated with hydrolysis of the nucleotide ATP to ADP and with the simultaneous structural change to a less stable actin polymer, as lateral contacts are lost in the less flexible helical structure.^{74,75} Thus, the variability in the helicity of the T3/H3 region seen in our simulations together with the fact that the T3 loop of β -tubulin has been observed to be destabilized upon hydrolysis of the bound GTP to GDP,⁶⁸ raise the possibility that upon hydrolysis of the GTP bound to β -tubulin, the T3/H3 region might undergo a loop-to-helix transition similar to that in actin, causing a large-scale shift in the entire MT structure. Experimental testing of this hypothesis could be performed by applying techniques which have previously been used to analyze small changes in protein helical structure, such as Trp fluorescence,⁷⁶ FRET (Forster resonance energy transfer) emission spectra,^{77,78} or EPR (electron paramagnetic resonance) data.⁷⁸ In this way, structural comparisons could be made of the T3/H3 region in MTs bound by different nucleotides or nucleotide-analogues.

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